

- 3 (a) (i) to-and-fro / backward and forward motion (between two limits) B1 [1]
- (ii) no energy loss or gain / no external force acting / constant energy / constant amplitude B1 [1]
- (iii) acceleration directed towards a fixed point B1
acceleration proportional to distance from the fixed point / displacement B1 [2]
- (b) acceleration is constant (magnitude) M1
so cannot be s.h.m. A1 [2]

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4 (a) ability to do work

P1